



DATA ANALYST 3: ADVANCED SQL PORTFOLIO



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Portfolio Outline



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Introduction

As a Data Analyst, I wanted to find out if factors such as: gender, ethnicity, education level, family income, family education make a difference in a these students' wage and amount of student tuition. I was provided with 4 datasets:

country_info

student_academic_info

student_family_details

student_personal_details

I found that inner joining these datasets helped to give me the cleanest dataset to work with.



Root Cause Analysis

Root Cause Analysis is the process used to find what the root cause/underlying problem is in order to figure out how to solve the problem.

- Why does high or low family income matter in the student's level of education?
- Why do we need to know if the student's mother or father attended college?
- Does the region play a part in average wage?
- Does gender play a role in tuition?
- Does a certain ethnicity pay higher tuition?

Insights

I used Inner Join to join the 4 datasets provided using the ID as the common data to create a new table called: reports_student_colleges.

```
1 CREATE TABLE reports_student_colleges AS|
2 SELECT * FROM county_info
3 INNER JOIN student_academic_info ON county_info.id = student_academic_info.id
4 INNER JOIN student_family_details ON county_info.id = student_family_details.id
5 INNER JOIN student_personal_details ON county_info.id = student_personal_details.id
```

I then selected all the data from my table to convert into a CSV file.

```
1 SELECT * FROM reports_student_colleges;
```

Wages by region:

```
1 SELECT wage, region FROM reports_student_colleges;
```

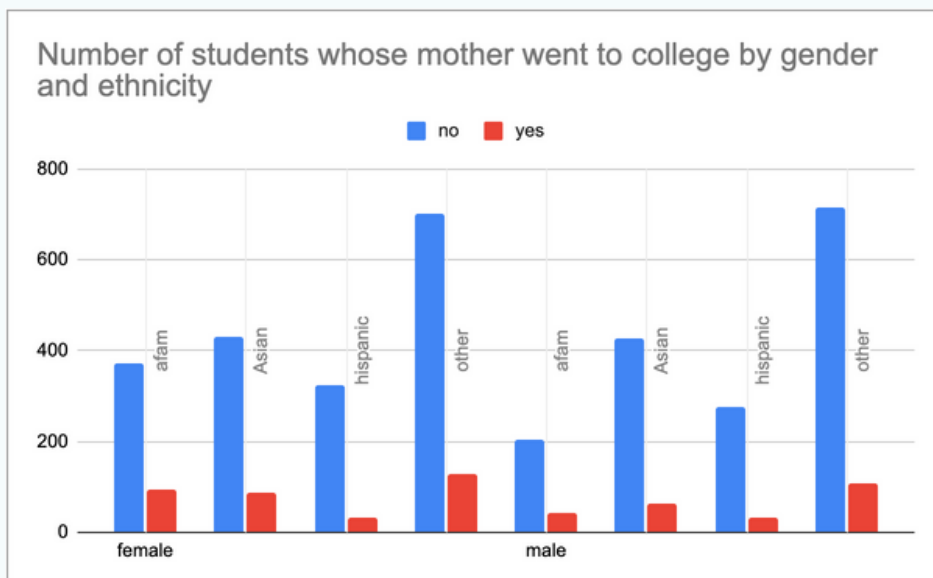
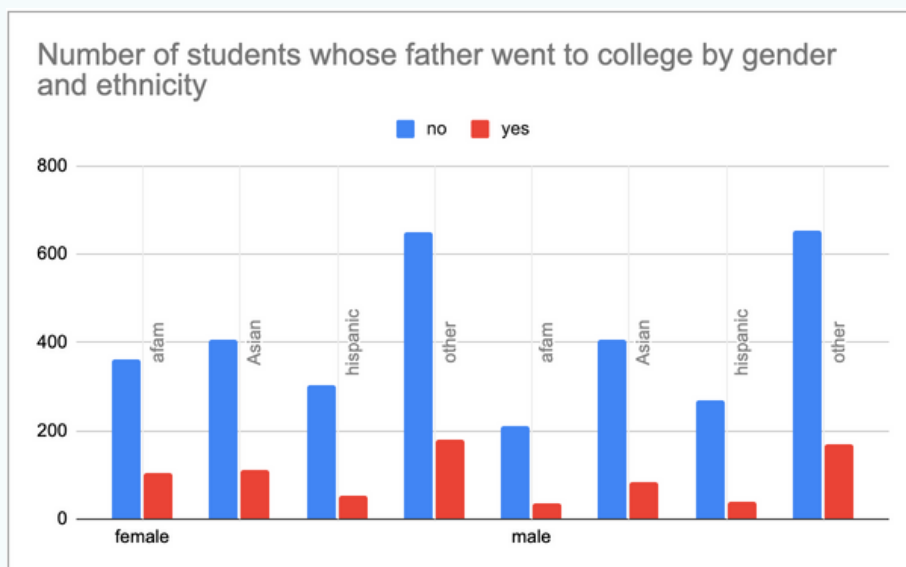
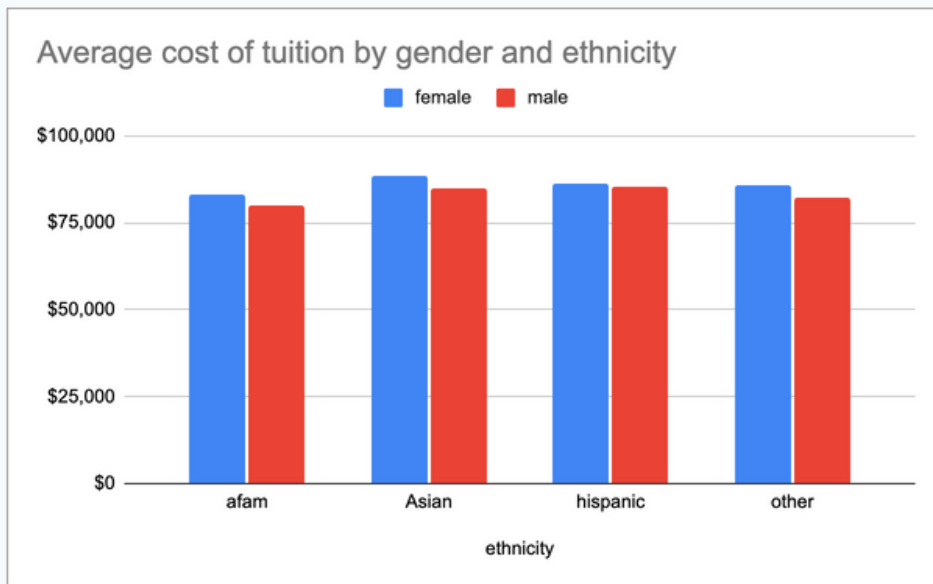
Number of females whose mother attended or did not attend college:

```
1 SELECT COUNT(gender), mcollege, ethnicity
2 FROM reports_student_colleges
3 WHERE gender = 'female'
4 GROUP BY mcollege;
```

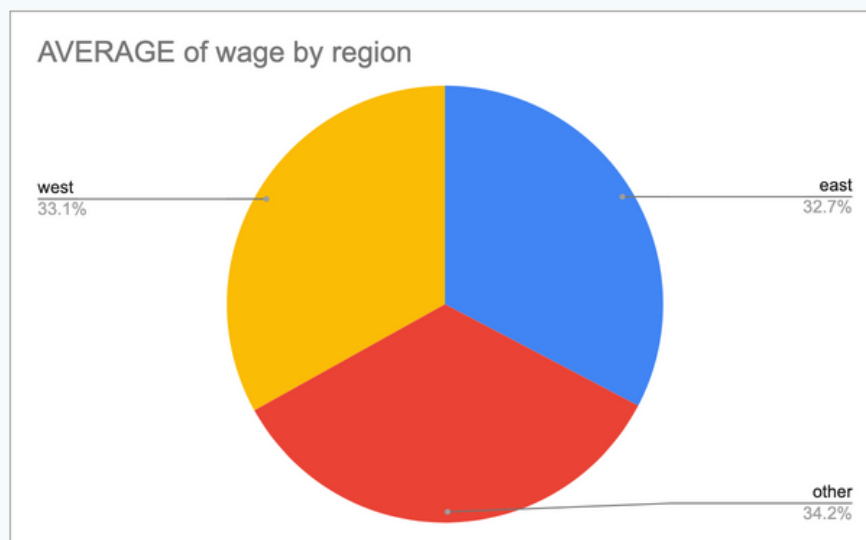
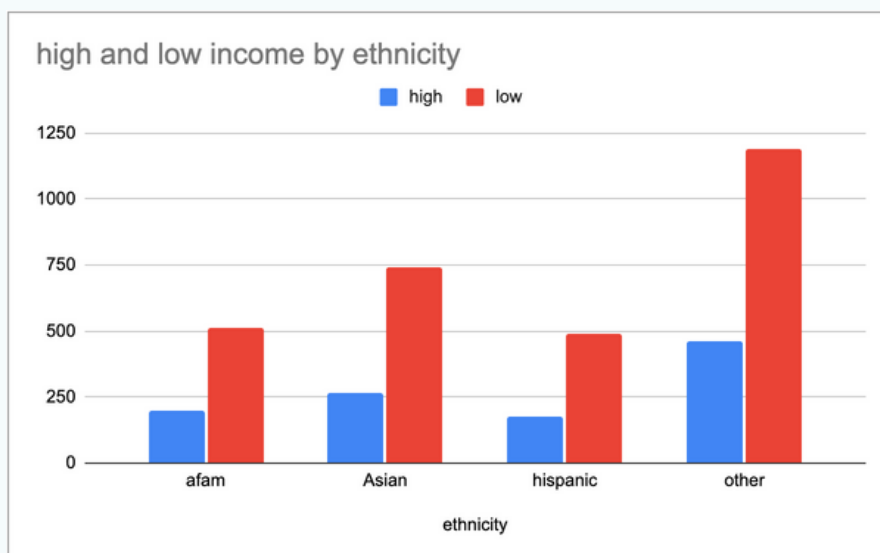
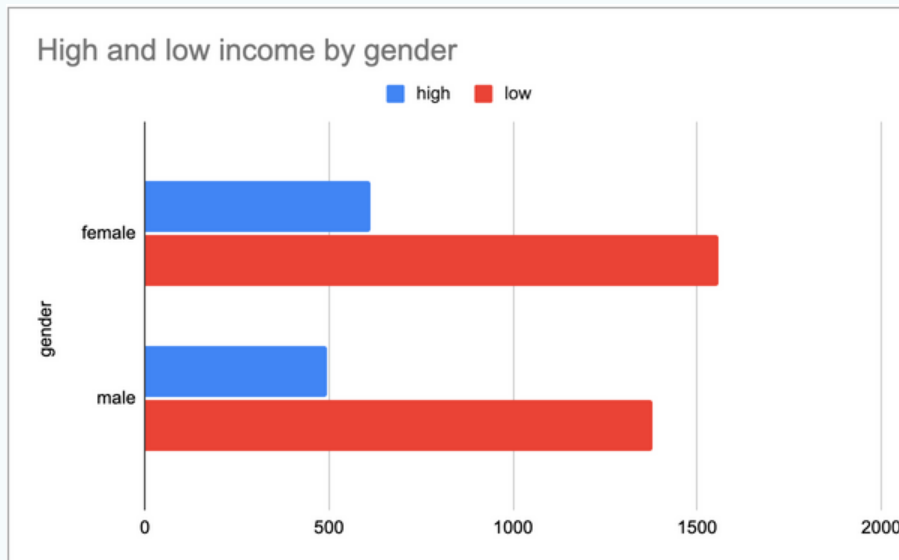
Number of males whose mother attended or did not attend college:

```
1 SELECT COUNT(gender), mcollege, ethnicity
2 FROM reports_student_colleges
3 WHERE gender = 'male'
4 GROUP BY mcollege;
```

Visualizations



Visualizations cont'd



Findings

Based on the data provided, I've found that Asian females and hispanic males pay higher tuition on average. Whereas, African America males and females pay lower tuition on average. This may be due to family income playing a role in financial aid or possibly the type of degree programs they are enrolling in.

For "Other" ethnicities, both male and female, it appears that their mother and fathers have not attended college. But, it also looks like they have the highest number of both parents how have attended. This is most likely because the "Other" categories may included more ethnicities.

I found that the average wage is pretty even across the east, west, and other regions.

I recommend that we dig a little further to find out what the degrees the students are obtaining. We should also be more clear about who the "Other" ethnicities are.

Conclusion

I've determined that Asians females pay more in tuition based on this dataset. I've also determined that both male and females have more fathers who have attended college than mothers.

In conclusion, I believe that we need more detailed information in order to determine why some ethnicities are paying higher or lower tuition. We also need to more info on what the "Other" ethnicities are as well as the other regions that were not listed.